

Grant Intelligence Platform

Transparent decision-support dashboard that ranks Northeast grant-funder prospects for EcoServants

Python

Pandas + NLP

100-point ranking

5/5 tests

PROJECT AT A GLANCE

Turns one or more IRS Business Master File CSVs into a cleaned, deduplicated, Northeast-only prospect list. It screens likely 501(c)(3) grantmakers, classifies mission alignment, computes TF-IDF cosine similarity to EcoServants, applies an explainable 100-point score, and serves search, filters, charts, a ranked table, and CSV download in Streamlit.

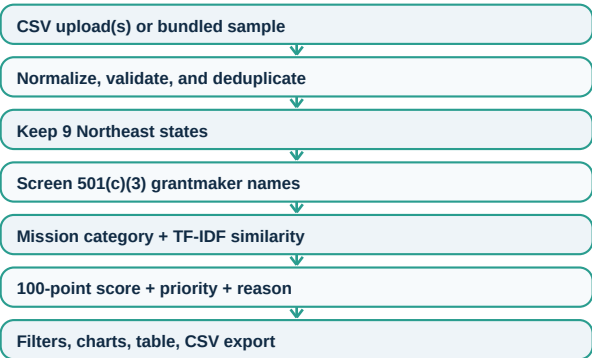
IMPACT AND VALUE

- Reduces a large nonprofit registry to a reviewable shortlist, focusing researcher time on stronger leads.
- Makes ranking auditable through component points and a plain-language reason for every prospect.
- Handles multi-file upload, normalization, invalid IDs, duplicate filings, missing text, and state scope.
- The included synthetic sample proves the full dashboard path without exposing real donor data.

LANGUAGES AND STACK

Languages/data: Python, TOML, CSV, Markdown.
Libraries: Pandas, scikit-learn, Streamlit, Plotly, pytest.
Methods: ETL validation, keyword classification, TF-IDF/cosine similarity, weighted scoring.

CODE FLOW - TOP TO BOTTOM



FILE MAP - 12 FILES

Inventory: all maintained project files; excludes .venv, .coverage, pytest/ruff caches, and generated exports.

.gitignore

Excludes environments, caches, raw/generated CSVs, exports, secrets, and editor state.

classification.py

Assigns mission categories by transparent keyword counts and records matched terms.

data/sample_bmf.csv

Synthetic 14-row IRS-style fixture with duplicates, invalid data, and varied missions.

LICENSE

Full MIT license for reuse, modification, distribution, and sale.

README.md

Detailed architecture, data rules, scoring model, dashboard use, limitations, and research workflow.

scoring.py

Calculates six point components, priority level, selection reason, and sorted ranking.

.streamlit/config.toml

Sets dashboard theme, 500 MB upload limit, and disables usage telemetry.

dashboard.py

Runs the pipeline and builds Streamlit metrics, filters, four Plotly charts, table, and download.

data_processing.py

Loads CSVs; normalizes headers, EINs, states, subsection/NTEE/financial codes; dedupes and scopes.

nlp_similarity.py

Vectorizes mission text with TF-IDF and adds 0-100 cosine similarity to EcoServants.

requirements.txt

Pins compatible ranges for Pandas, scikit-learn, Plotly, Streamlit, and pytest.

tests/test_pipeline.py

Tests cleaning, schema errors, categories, similarity/ranking, charts, and cached pipeline output.